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C9555 A Practical Guide to Simulations

Visualization

Ivo Kabelka

Common File Formats

- Fixed-width field file formats
- PDB
  - Legacy file format
  - Widely used
  - Coordinates in Ångströms
- GRO
  - Used by GROMACS
  - Coordinates in nanometers

PDB File Format – ATOM Record

1 - 6	Record name	"ATOM "	
7 - 11	Integer	serial	Atom serial number.
13 - 16	Atom	name	Atom name.
17	Character	altLoc	Alternate location indicator.
18 - 20	Residue name	resName	Residue name.
22	Character	chainID	Chain identifier.
23 - 26	Integer	resSeq	Residue sequence number.
27	AChar	iCode	Code for insertion of residues.
31 - 38	Real(8.3)	x	Orthogonal coordinates for X in Angstroms.
39 - 46	Real(8.3)	y	Orthogonal coordinates for Y in Angstroms.
47 - 54	Real(8.3)	z	Orthogonal coordinates for Z in Angstroms.
55 - 60	Real(6.2)	occupancy	Occupancy.
61 - 66	Real(6.2)	tempFactor	Temperature factor.
77 - 78	LString(2)	element	Element symbol, right-justified.
79 - 80	LString(2)	charge	Charge on the atom.

https://www.wwpdb.org/documentation/file-format

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PDB File Format – HETATM Record

1 - 6	Record name	"HETATM"	
7 - 11	Integer	serial	Atom serial number.
13 - 16	Atom	name	Atom name.
17	Character	altLoc	Alternate location indicator.
18 - 20	Residue name	resName	Residue name.
22	Character	chainID	Chain identifier.
23 - 26	Integer	resSeq	Residue sequence number.
27	AChar	iCode	Code for insertion of residues.
31 - 38	Real(8.3)	x	Orthogonal coordinates for X.
39 - 46	Real(8.3)	y	Orthogonal coordinates for Y.
47 - 54	Real(8.3)	z	Orthogonal coordinates for Z.
55 - 60	Real(6.2)	occupancy	Occupancy.
61 - 66	Real(6.2)	tempFactor	Temperature factor.
77 - 78	LString(2)	element	Element symbol; right-justified.
79 - 80	LString(2)	charge	Charge on the atom.

https://www.wwpdb.org/documentation/file-format

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PDB File Format – CRYST1 Record

1 - 6	Record name	"CRYST1"	
7 - 15	Real(9.3)	a	a (Angstroms).
16 - 24	Real(9.3)	b	b (Angstroms).
25 - 33	Real(9.3)	c	c (Angstroms).
34 - 40	Real(7.2)	alpha	alpha (degrees).
41 - 47	Real(7.2)	beta	beta (degrees).
48 - 54	Real(7.2)	gamma	gamma (degrees).
56 - 66	LString	sGroup	Space group.
67 - 70	Integer	z	Z value.

Also used to define the dimensions of a simulation box

https://www.wwpdb.org/documentation/file-format

PDB File Format – Example

```
ATOM   2932  N   SER A 305      39.108 204.720 141.274  1.00 91.76           N
ANISOU 2932  N   SER A 305    13467  10190  11207  -2275    542    -32       N
ATOM   2933  CA  SER A 305      40.420 204.614 141.903  1.00 92.59           C
ANISOU 2933  CA  SER A 305    13319  10465  11395  -2374    587      6       C
ATOM   2934  C   SER A 305      40.841 203.133 141.985  1.00 93.18           C
ANISOU 2934  C   SER A 305    13167  10661  11576  -2230    641     27       C
ATOM   2935  O   SER A 305      40.091 202.279 142.474  1.00 93.31           O
ANISOU 2935  O   SER A 305    13129  10683  11640  -2084    559     14       O
ATOM   2936  CB  SER A 305      40.381 205.235 143.300  1.00 94.14           C
ANISOU 2936  CB  SER A 305    13464  10697  11609  -2448    452      4       C
ATOM   2937  OG  SER A 305      41.530 204.957 144.084  1.00 96.52           O
ANISOU 2937  OG  SER A 305    13505  11170  11998  -2526    459     41       O
TER    2938      SER A 305
HETATM 2939  C10 RVZ A1201      21.615 178.722 159.314  1.00 60.42           C
ANISOU 2939  C10 RVZ A1201     8603   6116   8238   -116   -312    -40       C
HETATM 2940  C13 RVZ A1201      22.505 173.854 157.851  1.00 61.58           C
ANISOU 2940  C13 RVZ A1201     9106   6040   8250     -7   -171    -32       C
HETATM 2941  C21 RVZ A1201      21.175 170.896 153.227  1.00 67.05           C
ANISOU 2941  C21 RVZ A1201    10349   6377   8750   -176   -159    -91       C
HETATM 2942  C22 RVZ A1201      20.727 169.667 153.591  1.00 67.06           C
ANISOU 2942  C22 RVZ A1201    10460   6300   8721   -194   -141    -85       C
```


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GRO File Format

1 – 5	5	**resSeq**	residue number
6 – 10	5	**resName**	left-justified in writer (%-5s)
11 – 15	5	**name**	right-justified (%5s)
16 – 20	5	**serial**	atom number
21 – 28	8.3	**x**	coordinate
29 – 36	8.3	**y**	coordinate
37 – 44	8.3	**z**	coordinate
45 – 52	8.4	**vx**	velocity (optional)
53 – 60	8.4	**vy**	velocity (optional)
61 – 68	8.4	**vz**	velocity (optional)

GRO File Format – Example

Comment

Number of atoms

```
Martini system t= 8175728.00000 step= 272524275
26047
    1SER     BB    1   49.641   3.346   7.145  0.1125 -0.3296  0.1017
    1SER    SC1    2   49.717   3.526   7.242 -0.1303  0.2001  0.2547
    2LEU     BB    3   49.506   3.258   6.879  0.2906 -0.0745 -0.0806
    2LEU    SC1    4   49.319   3.384   6.670 -0.2321 -0.2367 -0.0223
    3ALA     BB    5   49.448   2.969   6.976 -0.1043  0.0199 -0.0125
    4LYS     BB    6   49.282   3.037   7.229  0.0012  0.1002  0.0363
    4LYS    SC1    7   49.404   2.888   7.484  0.0026  0.1613 -0.1663
    4LYS    SC2    8   49.701   2.901   7.483  0.1475  0.1196 -0.0204

...

17012ION     NA26039  41.607   3.367   7.286 -0.0903  0.5455 -0.0379
17013ION     NA26040  32.340   3.151   3.446  0.3649  0.0136  0.0702
17014ION     NA26041  62.905   0.871   9.095  0.3234 -0.1576 -0.1059
17015ION     NA26042  43.507   0.122   4.306  0.2064 -0.2798  0.0895
17016ION     NA26043  20.403   0.016   8.819  0.0529  0.3402  0.1790
17017ION     NA26044  31.448   3.885   1.996  0.1403  0.1520 -0.0362
17018ION     NA26045  24.151   0.994   8.729  0.0762  0.2261 -0.1425
17019ION     NA26046  44.115   2.095   3.824  0.0155  0.2877 -0.0153
17020ION     NA26047  51.048   4.040   3.213  0.0862  0.3467  0.1503
 63.32880   4.20000  11.11281
```

Box dimensions

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PyMOL

- ctrl + alt + t  Shortcut to open a terminal window
- Start a PyMOL session:

```
$ module add pymol
$ pymol
```
bash

- Download protein structures from the PDB database:

```
fetch 2YDV 5MZP 7ARO
```
pymol

Each protein has a unique four-letter identifier
Each small molecule has a unique three-letter identifier

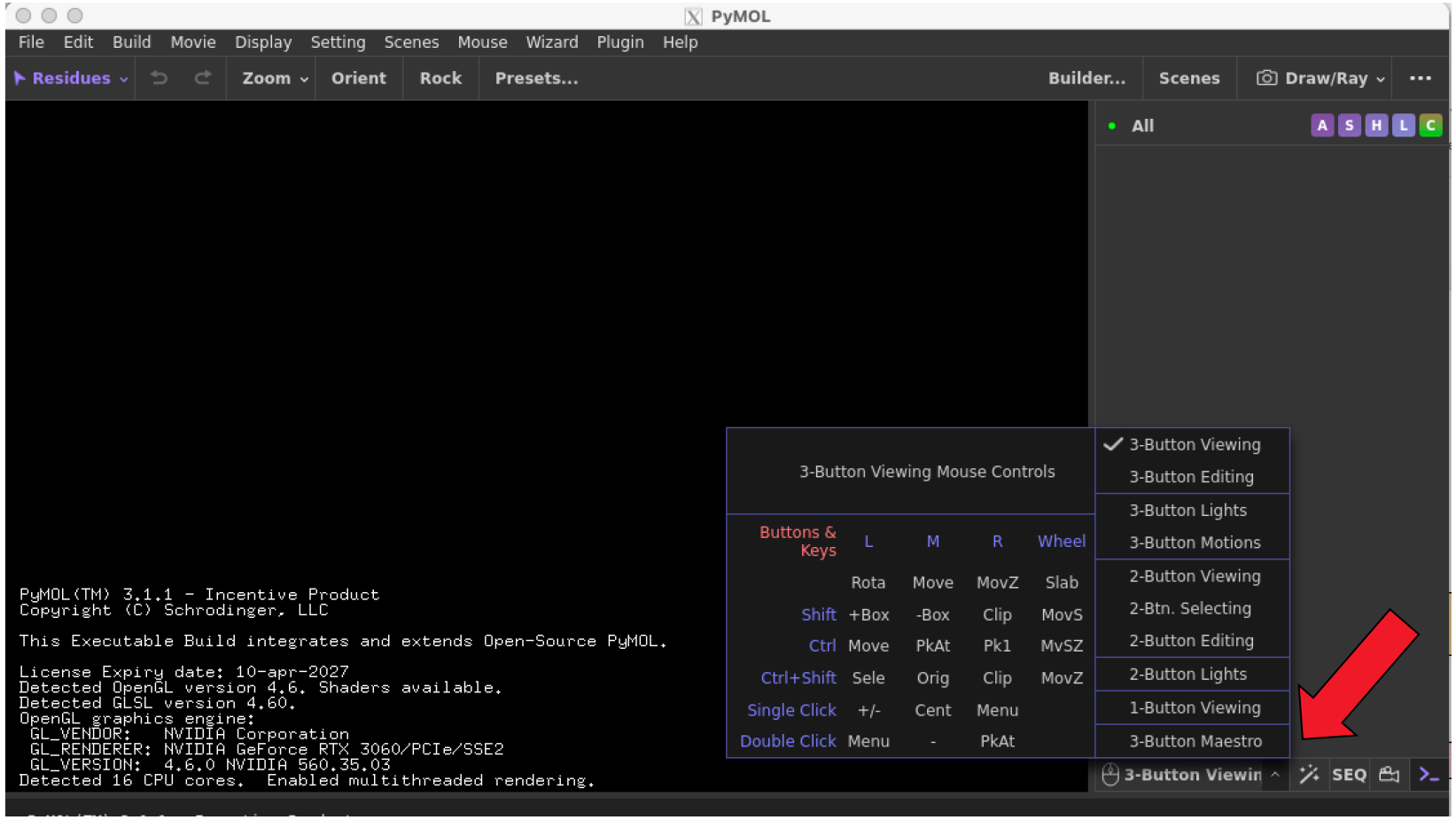
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PDB File – Contents

https://www.rcsb.org/structure/7ARO

https://files.rcsb.org/view/7ARO.pdb

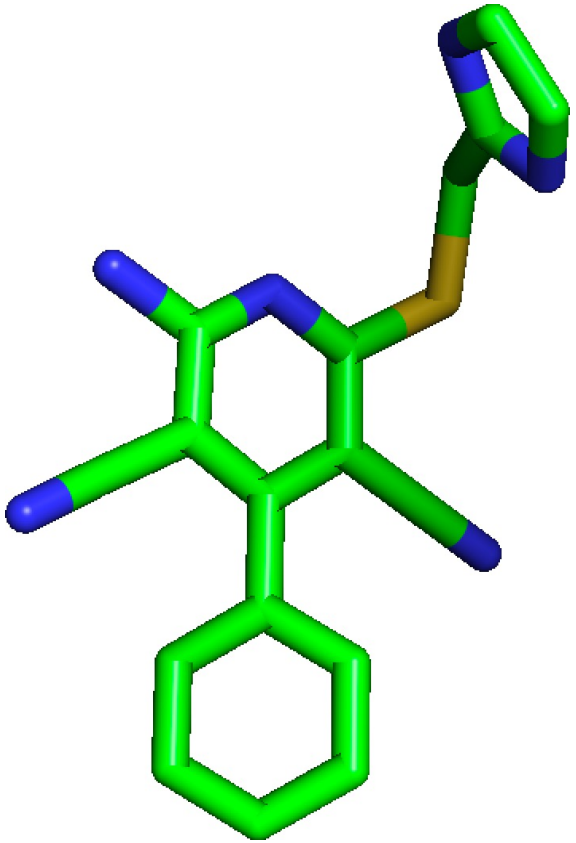
PyMOL – Navigation



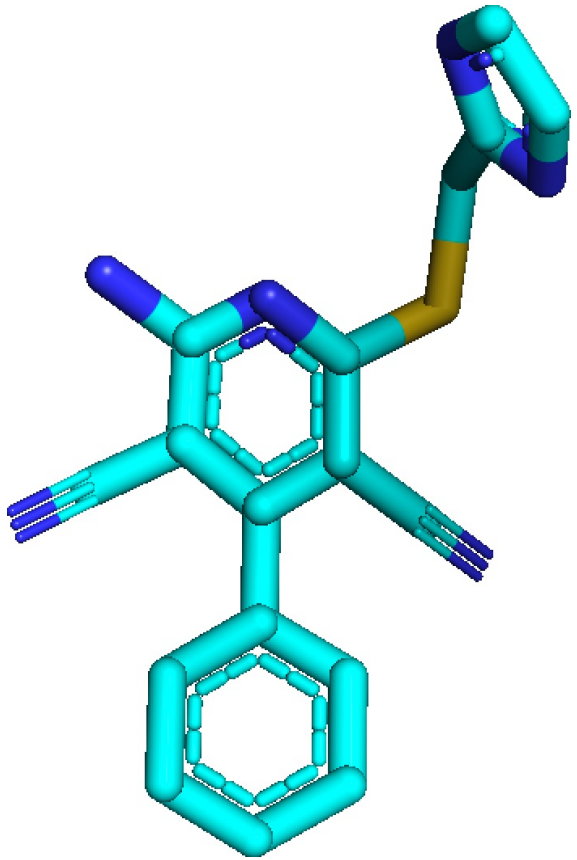
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PyMOL – ATOM / HETATM

ATOM

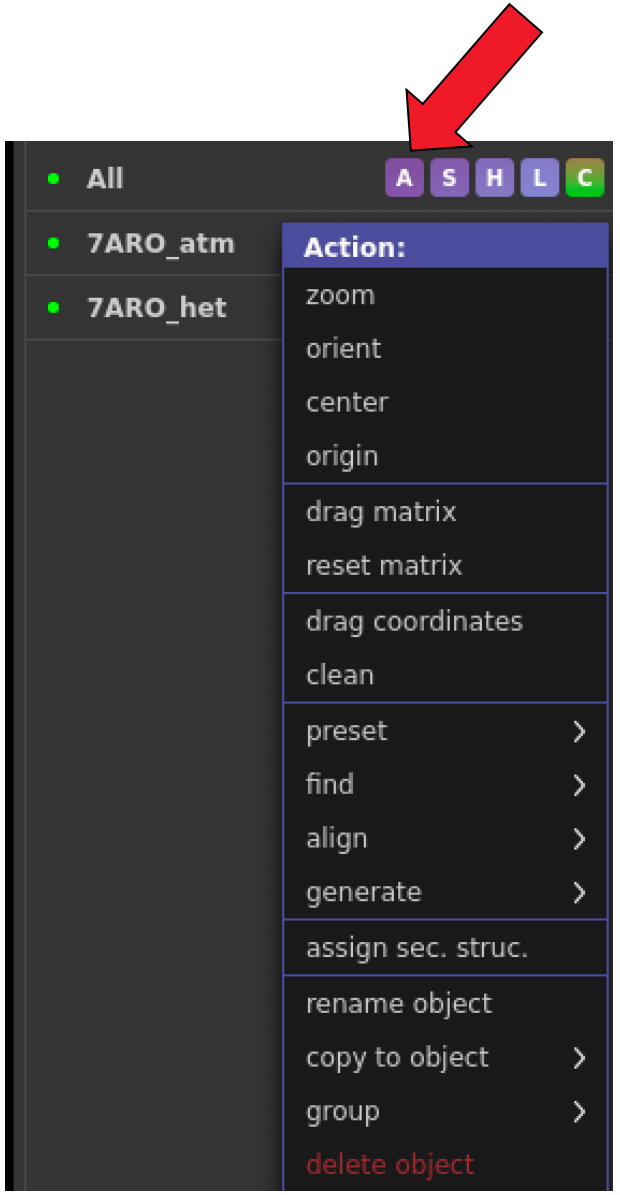


HETATM



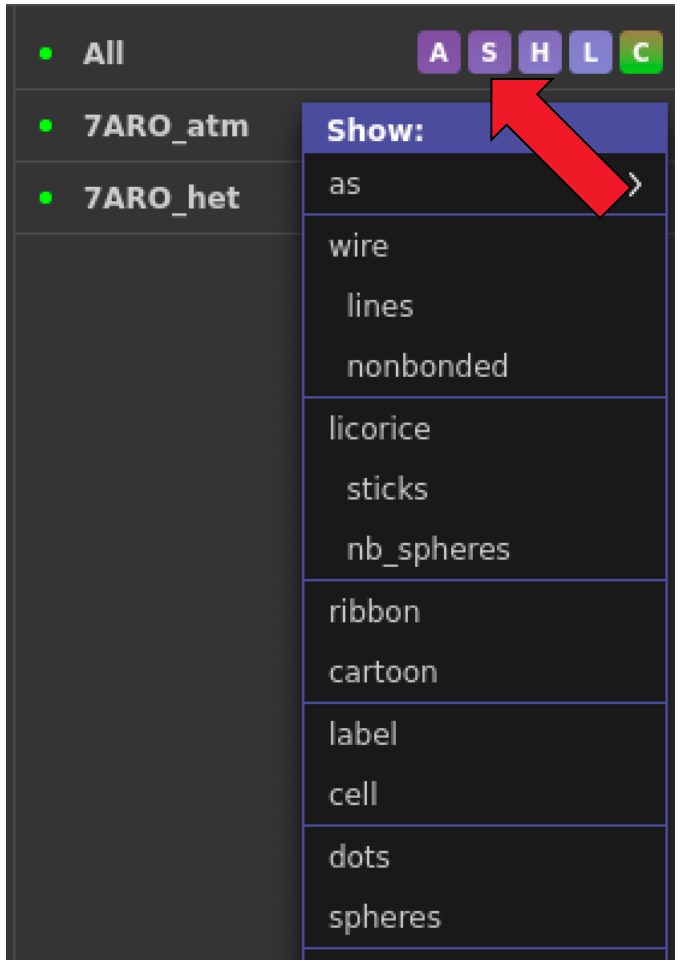
PyMOL – Actions

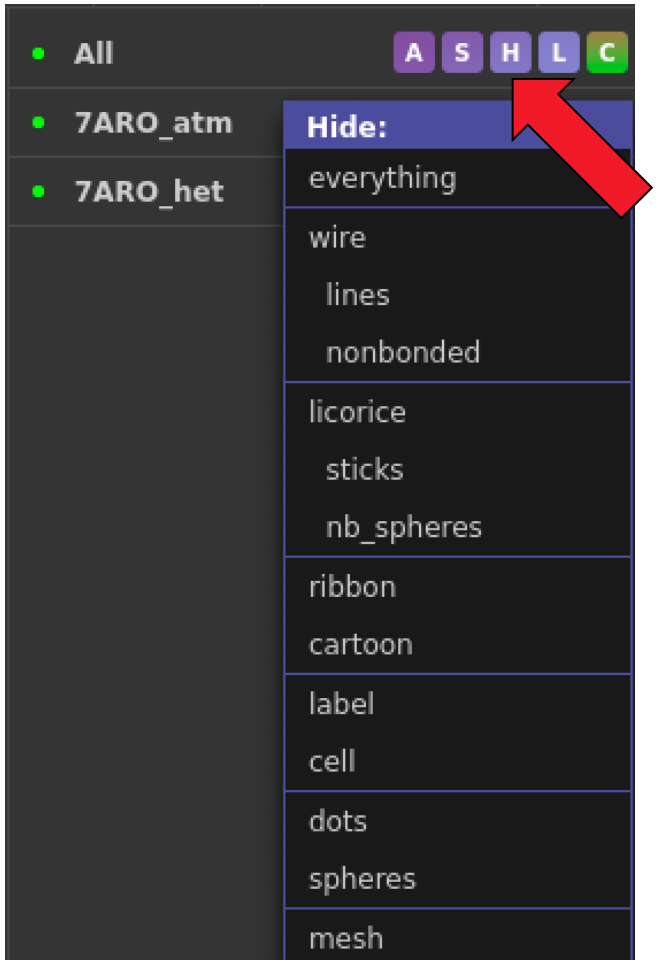
- Adjust the view
  - Zoom, center
- Align objects
  - Superimpose structures with align
- Delete atoms
  - Remove selected atoms
- Add/remove hydrogens
  - Modify hydrogen atoms

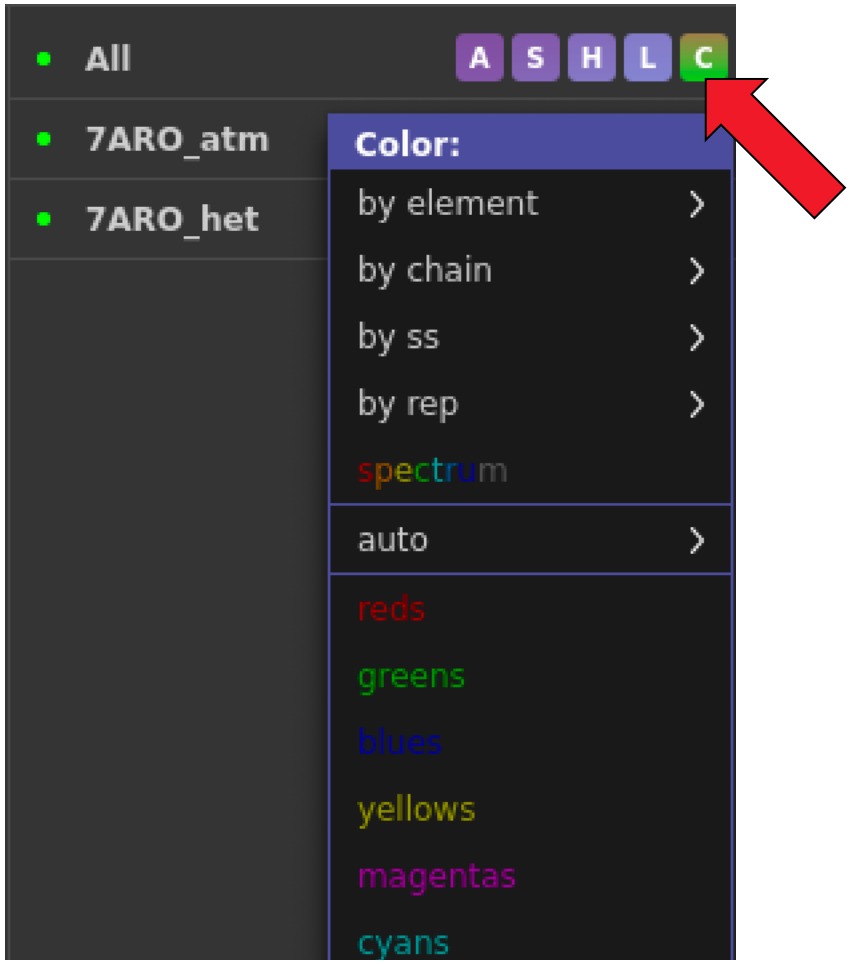


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PyMOL – Show / Hide / Colors





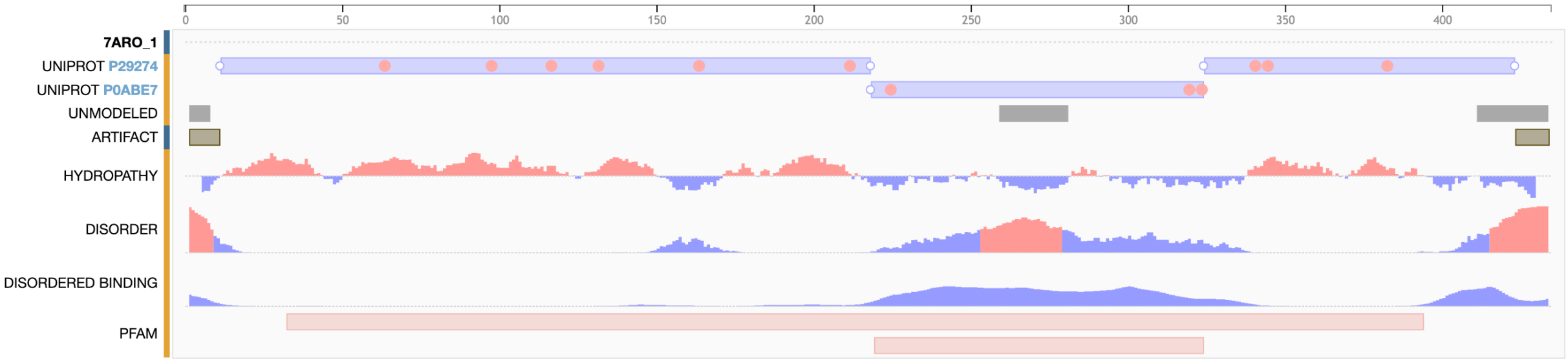


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PDB File – Problems and Considerations

- Missing atoms
- Missing residues
- Missing hydrogens (protonation / tautomeric states)
- Co-crystallized atoms/molecules
  - Crystal water molecules
- Alternative locations
- Mutations
- Engineered domains
- Post-translational modifications as separate entities
- Inconsistent (residue) numbering and chain labels
  - E.g. numbering of engineered domains, chain A as a first chain or R to label a receptor

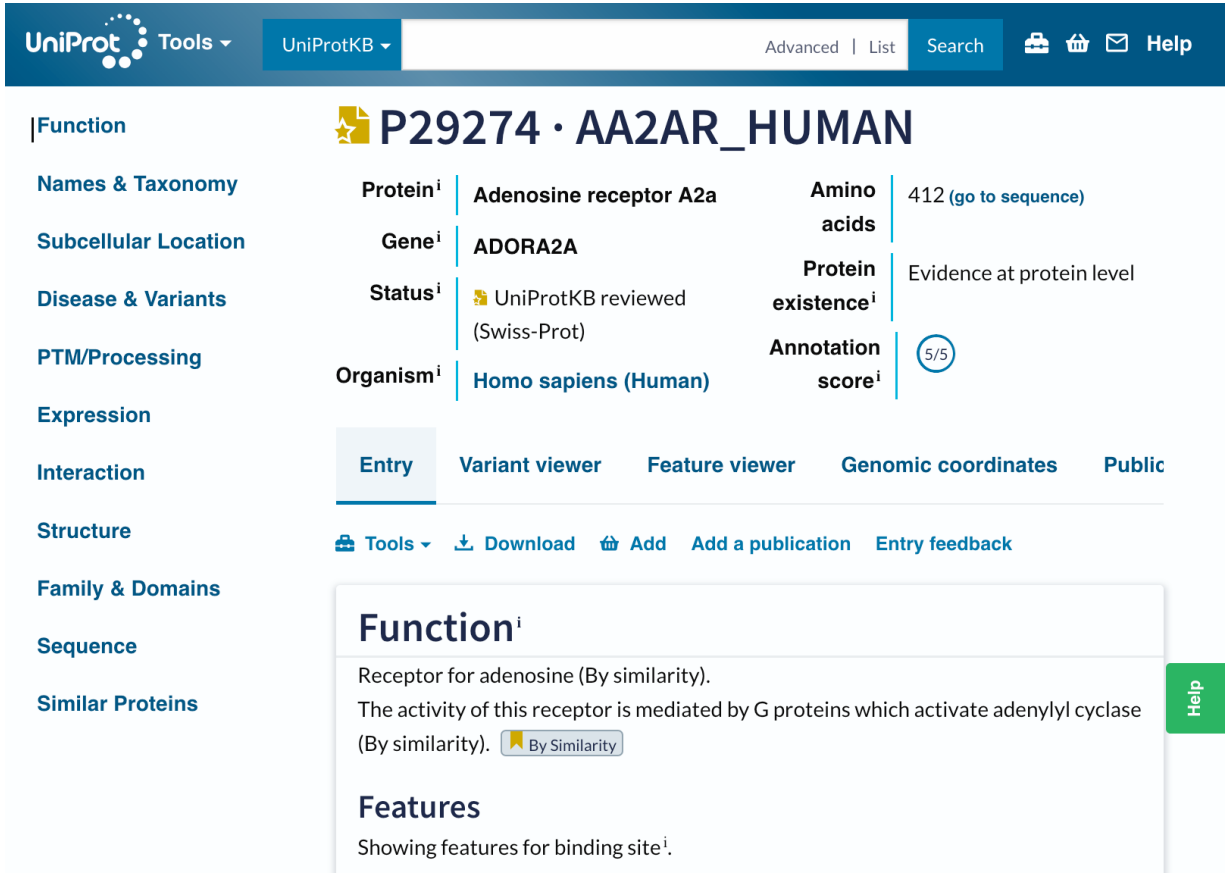
RCSB – Sequence Annotations



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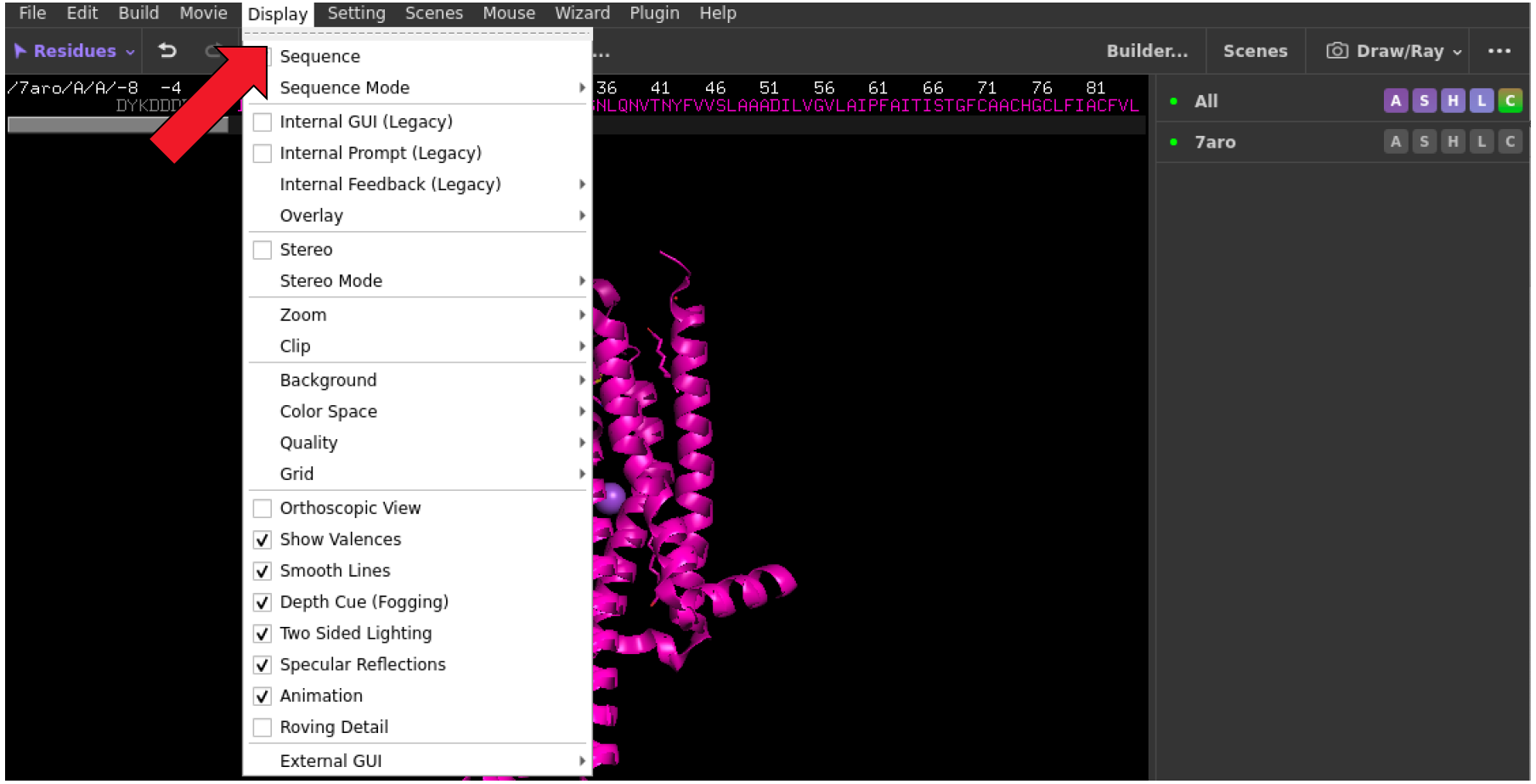
UniProt

- Protein annotations, canonical sequences, variants, etc.
- Adenosine receptor A2a
  - https://www.uniprot.org/uniprotkb/P29274/entry



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PyMOL – Sequence



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PyMOL – Selections

- https://pymolwiki.org/index.php/Selection_Algebra

```pymol
select sele, polymer.protein

select sele, name CA
```

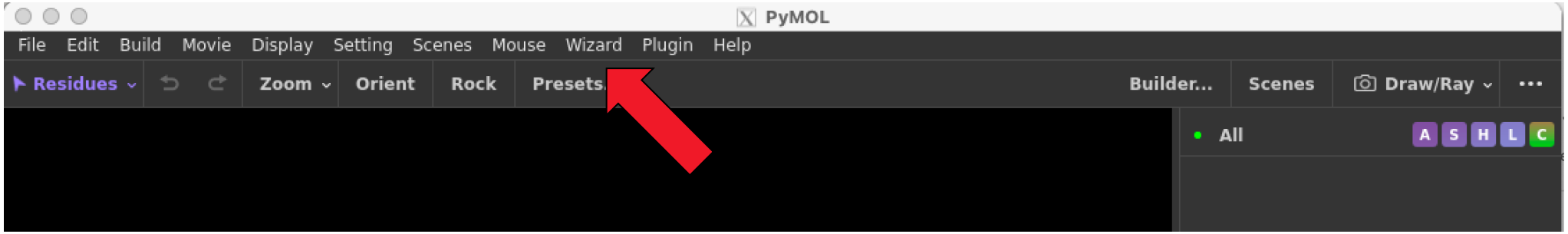
- **Task**
  - Hide everything
  - Select residues 168, 169, 246, 250, 253 and residue RVZ
  - Show selection as licorice

PyMOL – Atom Manipulation

- **Task**
  - Select co-crystallized molecules (try both selection algebra and “sequence”)
  - Remove co-crystallized molecules
  - Add polar hydrogens

PyMOL – Residue Manipulation

- Mutageneis wizard
  - Introduce or revert mutations (e.g., phosphomimetics)
  - Pick a specific rotamer
  - Add missing atoms (mutate a residue back to itself, i.e., Lys to Lys)
  - Add terminal capping (e.g., remove charges at chain breaks)

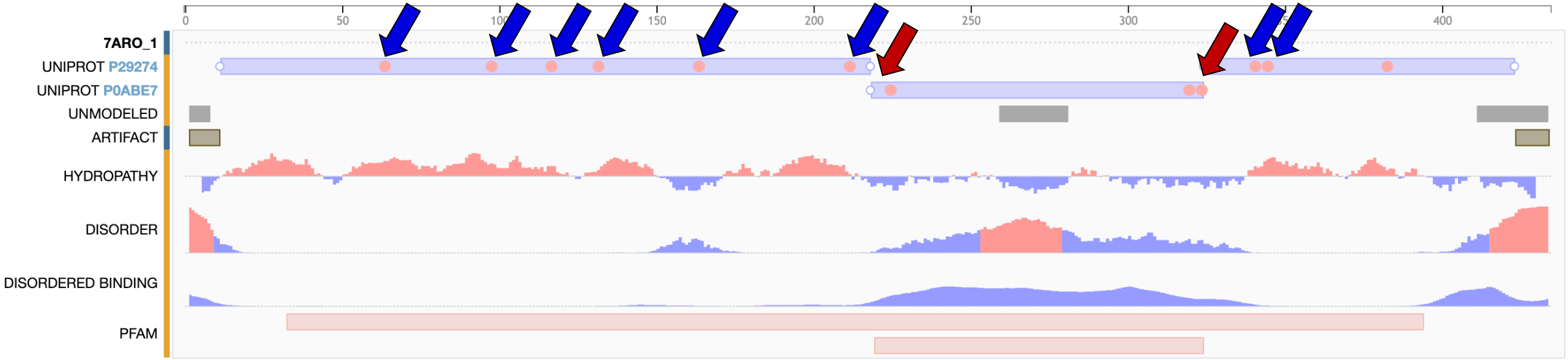


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PyMOL – Residue Manipulation

– Task

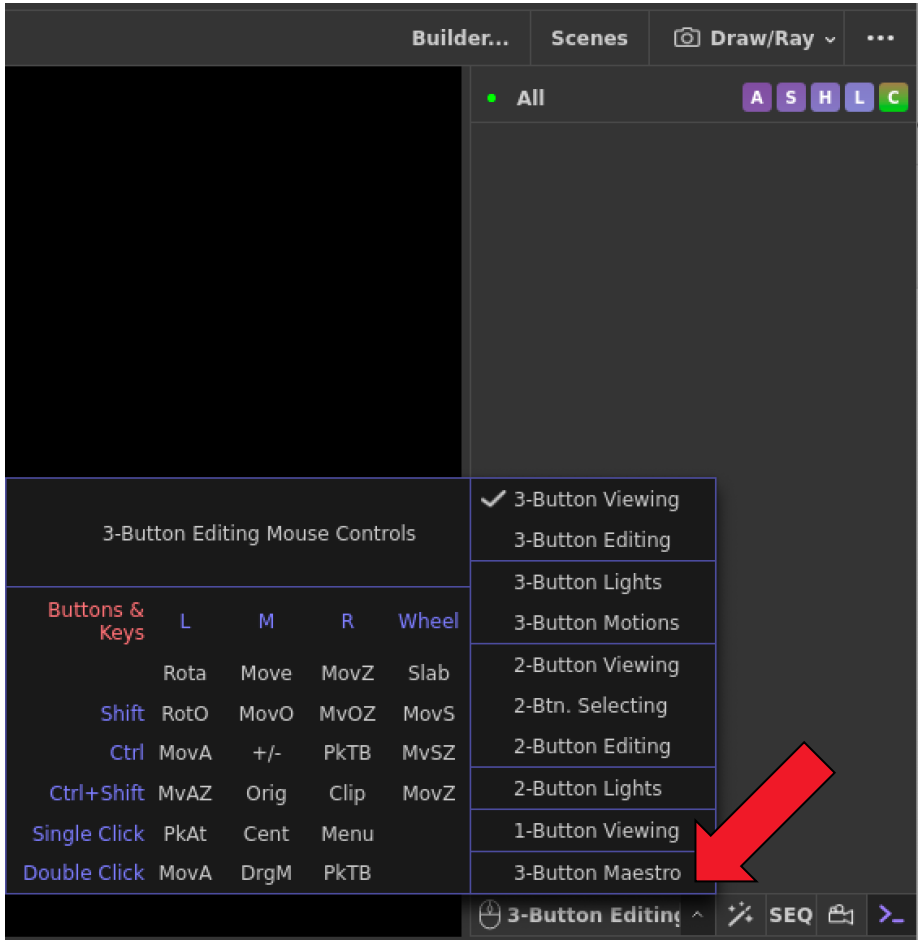
- Revert mutations
- Add terminal capping at chain breaks



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PyMOL – Molecule Manipulation

- **Task**
  - Switch to 3-Button Editing
  - Align structures of 2YDV and 7ARO
  - Hide everythong
  - Show the protein structures as cartoon
  - Show residue 169 (glutamic acid) as licorice
  - Change the torsion angles to match the rotamers



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PyMOL – fab

- Quickly generate peptides from sequence
  - https://pymolwiki.org/index.php/Fab

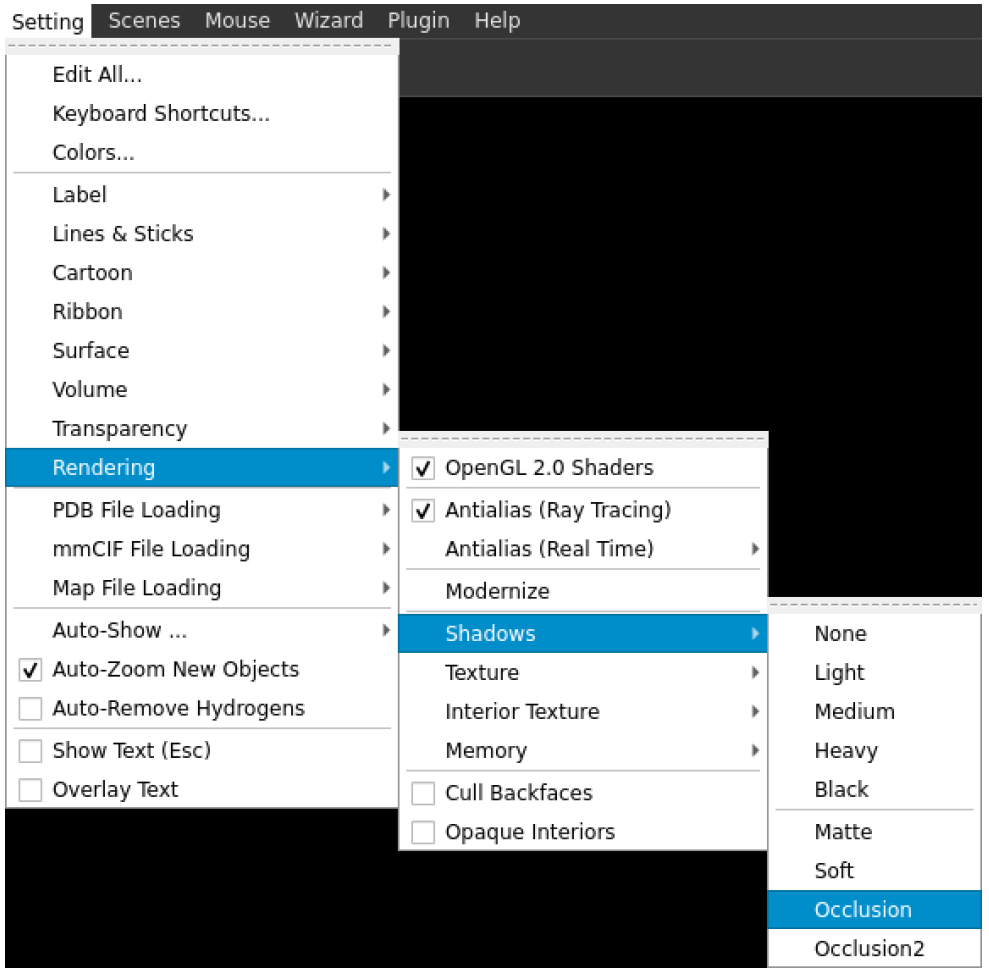
- **Task**
  - Generate a short α-helical peptide

PyMOL – Save Session and Logging

- Save session
  - Saves the current state (including the loaded structures)
  - Can be shared with collaborators
  - Useful for reproducibiliy
- Logging
  - Saves the user commands and actions to a file
  - Can be conveniently used to make scripts

PyMOL – Making Pictures

- PyMOL supports ray tracing
  - Shadows and ambient occlusion
  - Gives the structure depth
  - Pictures can be rendered on a transparent background
- **Task**
  - Make a publication-ready picture



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VMD

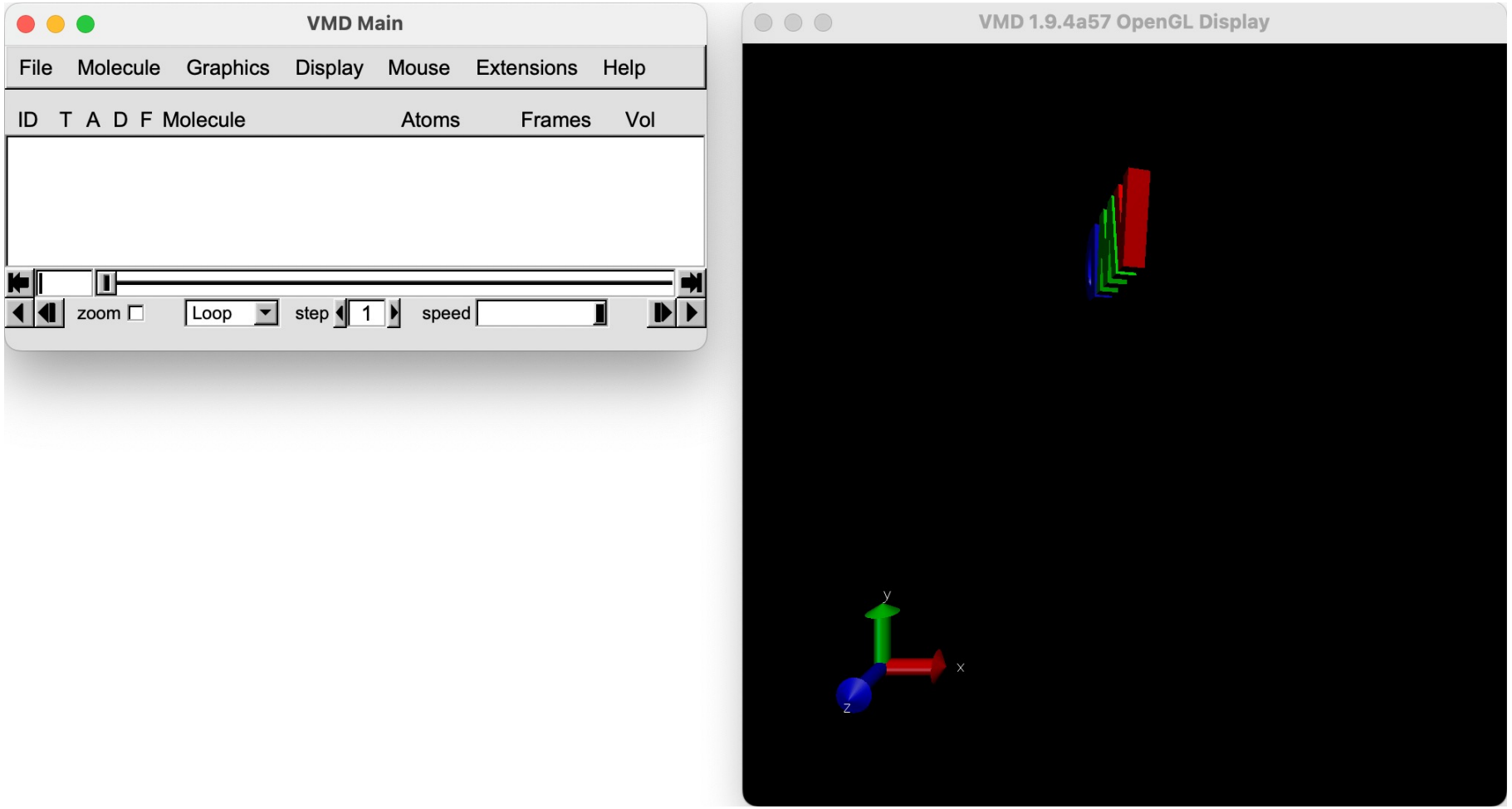
- ctrl + alt + t  Shortcut to open a terminal window
- Start a VMD session:

```
$ module add vmd
$ vmd
$ vmd snapshot.gro trajectory.xtc # GRO file contains topology information
$ vmd -m molecule1.pdb molecule2.pdb molecule3.pdb
```

bash

By default, multiple files are loaded as a trajectory
The number of atoms in a file has to match to add / append a trajectory

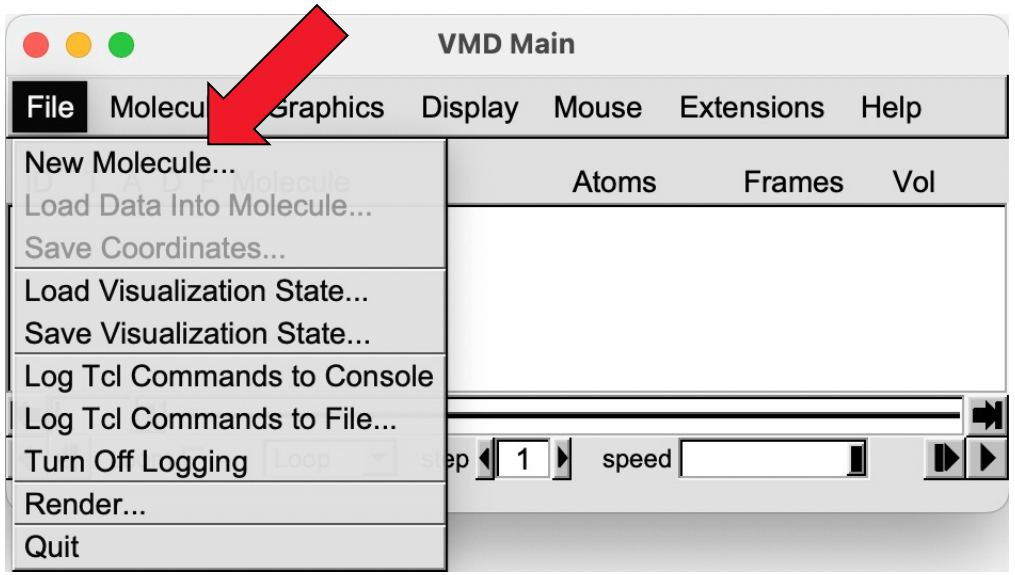
VMD – Interface



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VMD – Interface

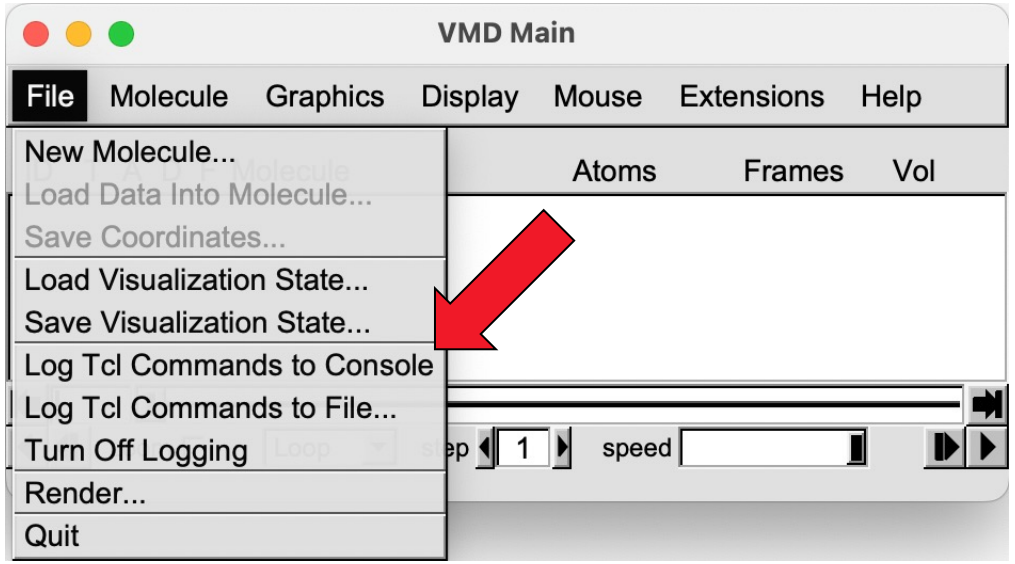
- Load a molecule
- Append a trajectory
  - Faster if loaded at once in background
- Load data into molecule



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VMD – Interface

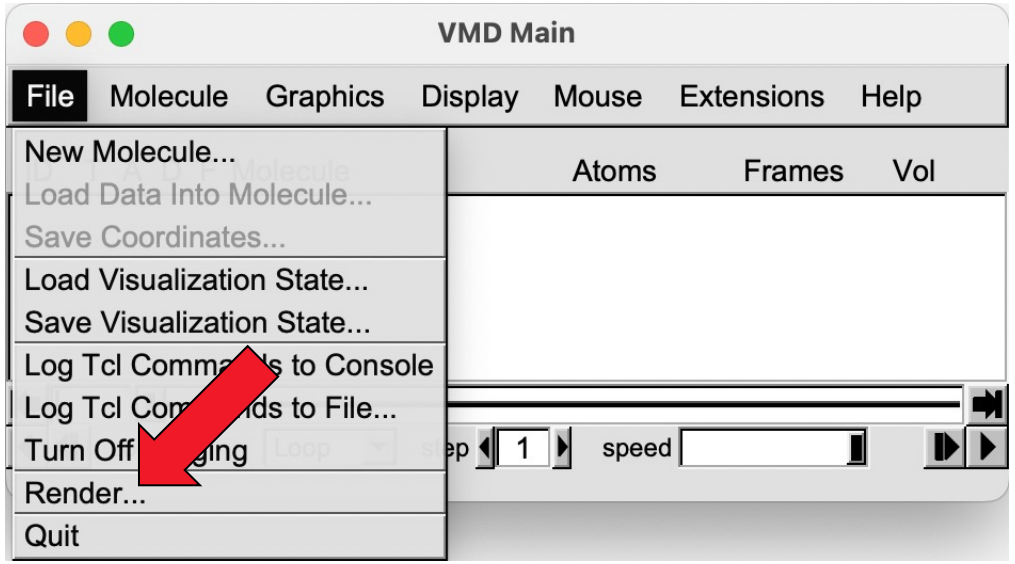
- Log Tcl commands
  - Useful to make scripts



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VMD – Interface

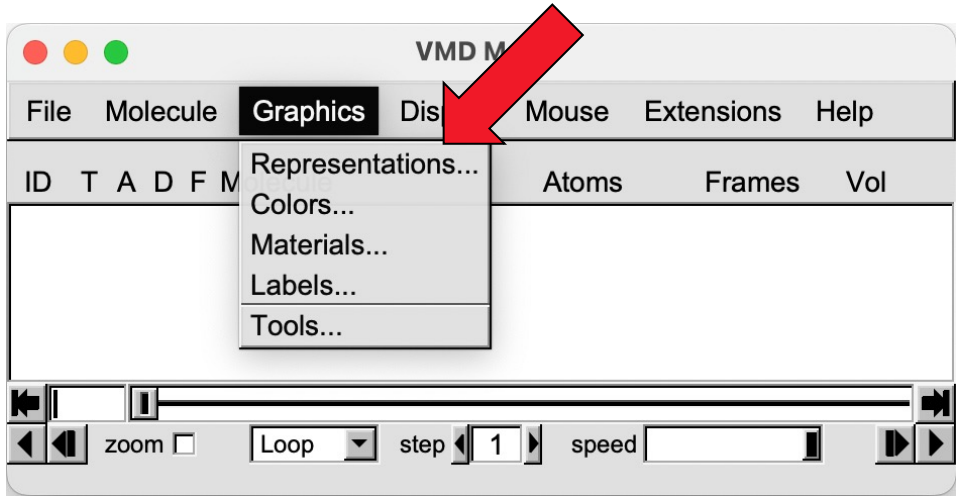
- Render...
  - Use tachyon for ray tracing
  - Faster if GPU-accelerated
  - Can also export 3D structure (obj), e.g., for 3D printing



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VMD – Interface

- Representations
  - Select what to display and how
  - Styles (cartoon, licorice, surfaces) and colors
  - Trajectory smoothing (frame averaging)
  - Show perodic images

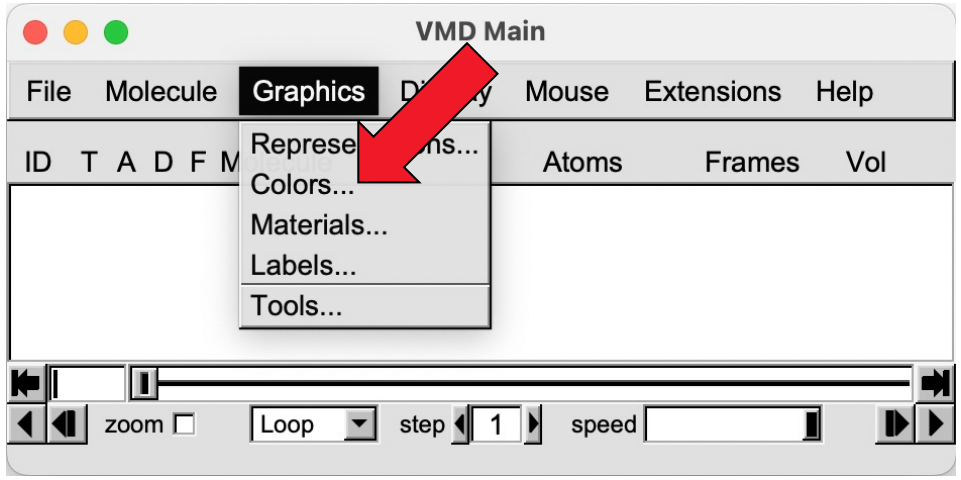


“AO” materials are great for ray tracing

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VMD – Interface

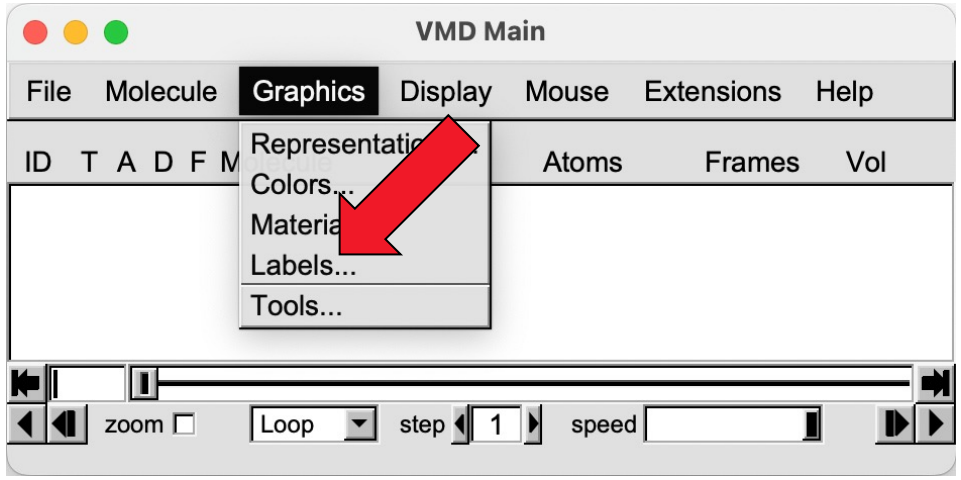
- Colors
  - Change the background color
  - Change colors of elements, residue types, etc.
  - Change the color palettes



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VMD – Interface

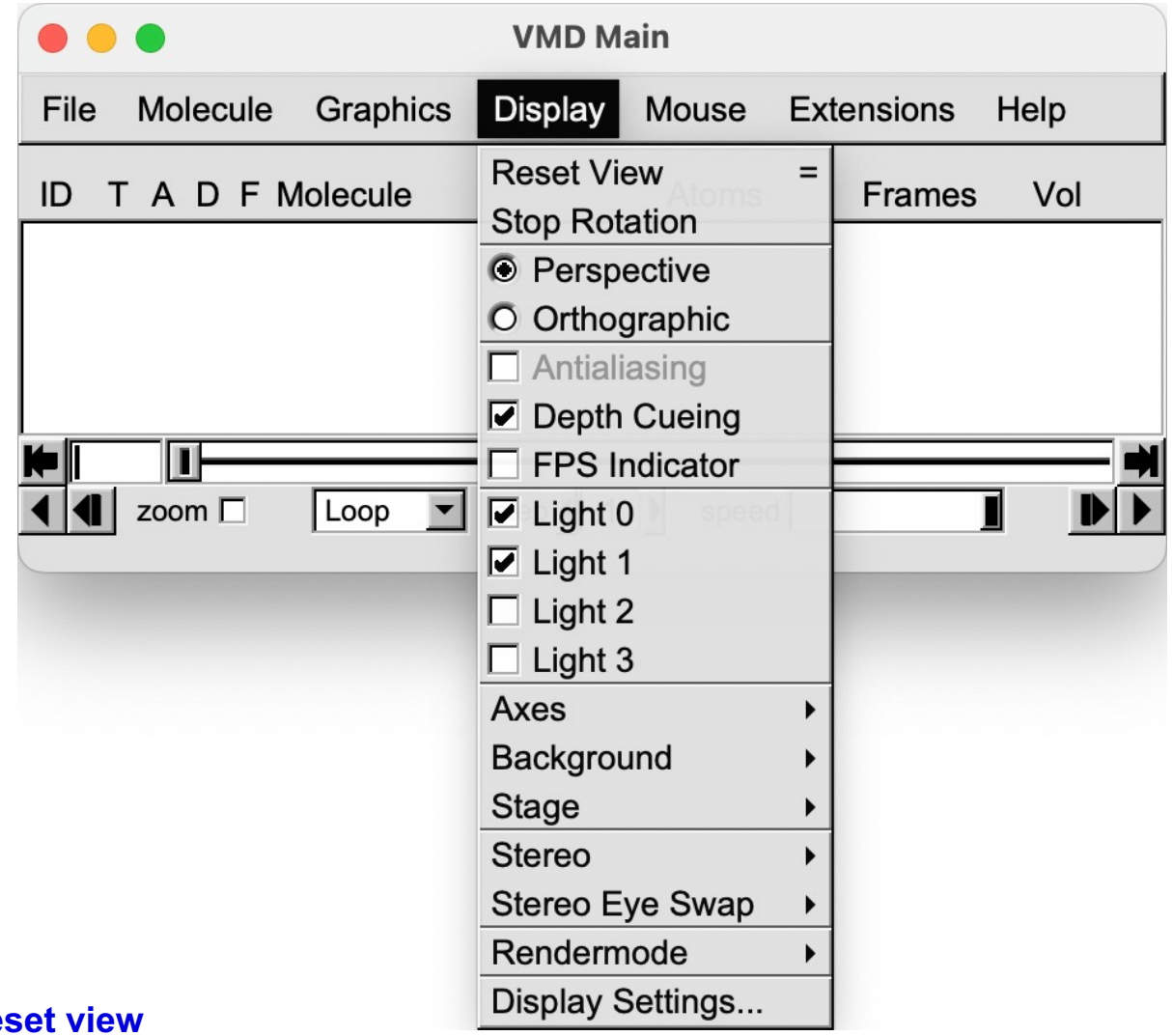
- Labels
  - Manage labels (atom names, distanes, angles, etc.)
  - Plot distances etc.



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VMD – Interface

- Display
  - Mostly self-explanatory
  - Depth cueing (fog that covers distant objects)
  - Rendermode (if supported by hardware, enables ray traced visualization)
  - Display settings (enable ambient occlusion and shadows)

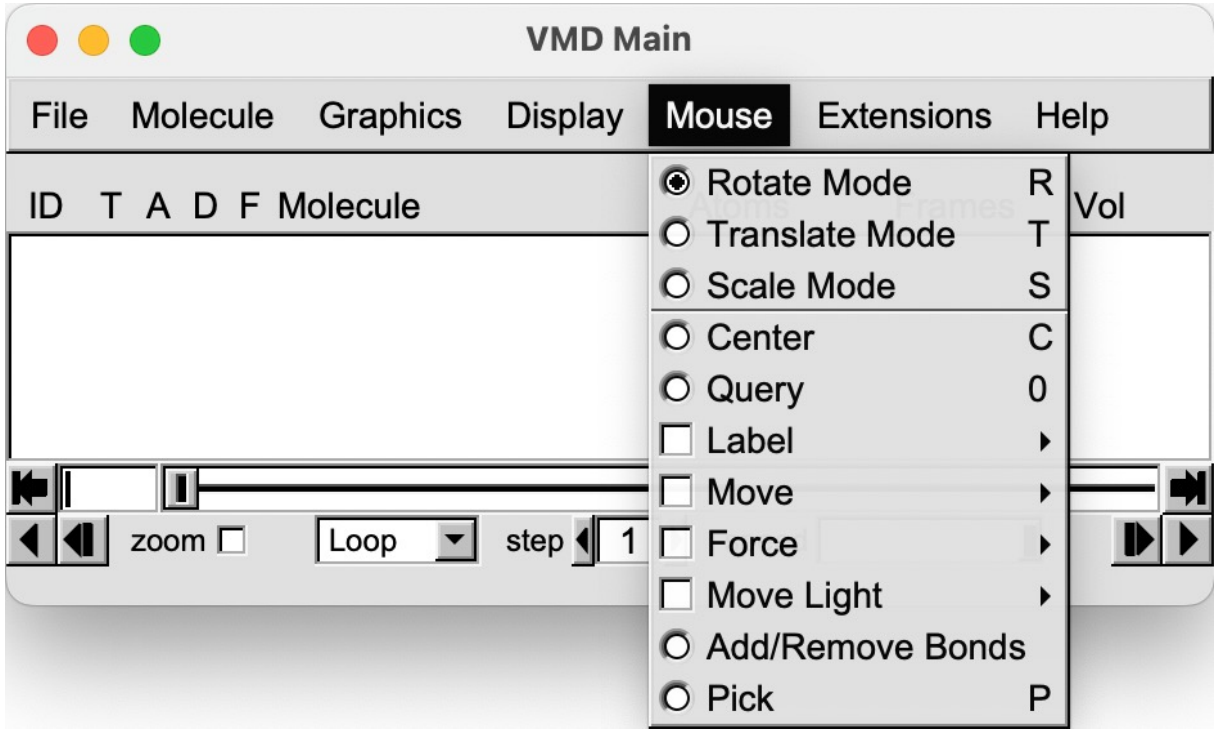


"=" symbol to reset view
"=" Centers the view on the visible representation

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VMD – Interface

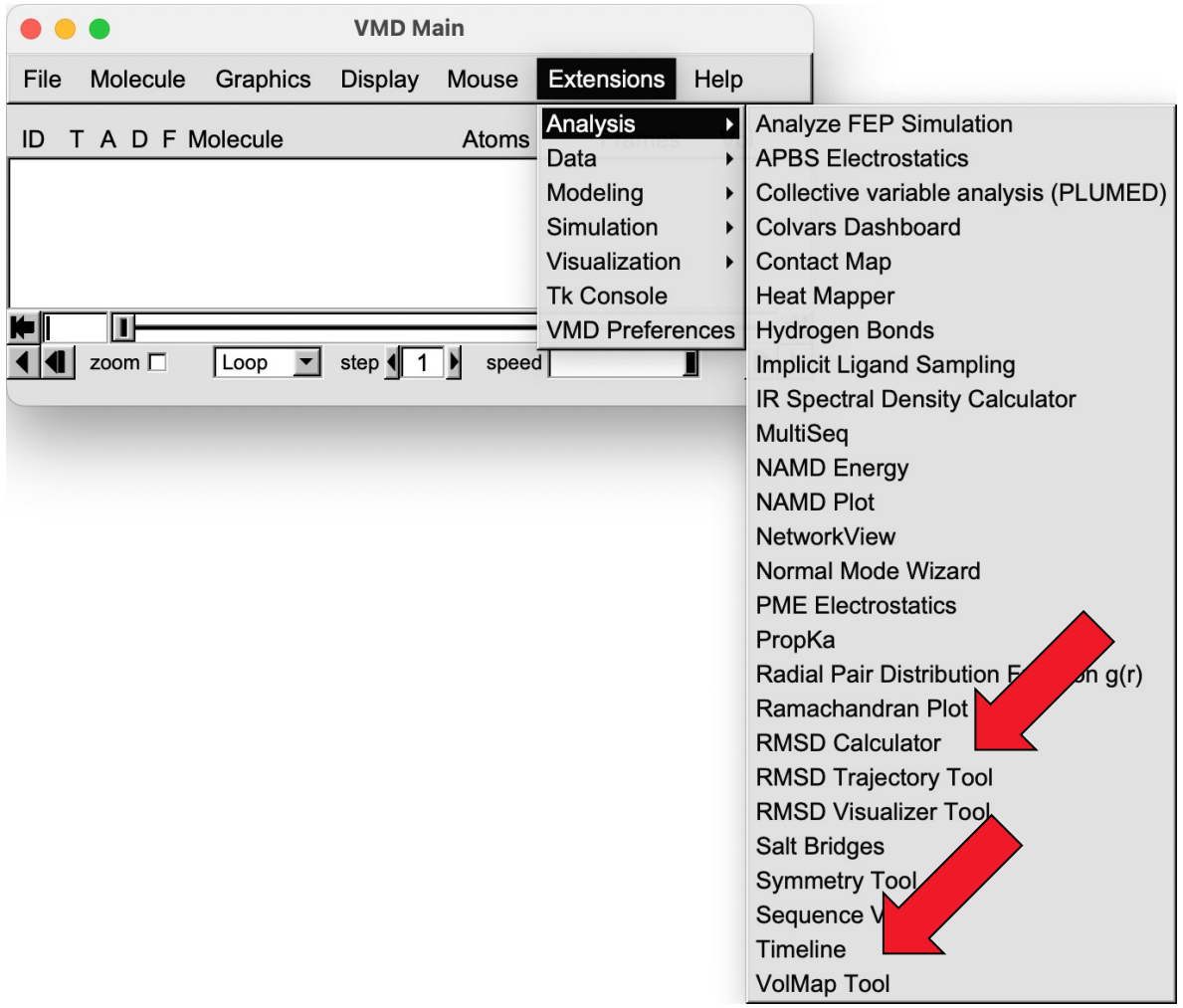
- Mouse
  - Change mouse behavior (shortcuts: R, T, S)
- Label
  - Label atoms
  - Measure distances, angles, etc.
- Move
  - Atom / molecule manipulation



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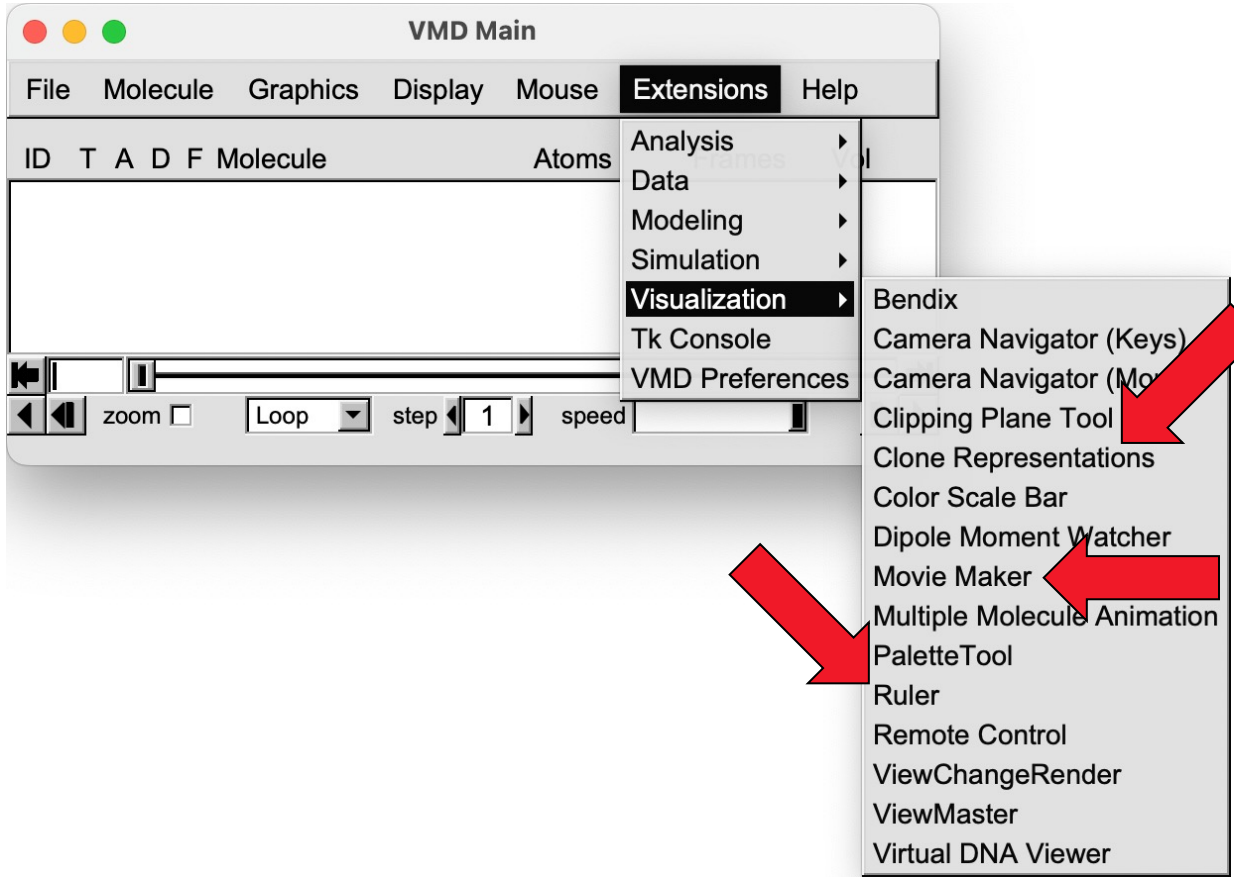
VMD – Interface

- RMSD
  - Align molecules / trajectory (and calculate RMSD)
- VolMap Tool
  - Needs alignment
  - Create 3D volumetric maps (find structural waters, coordinating ions, etc.)



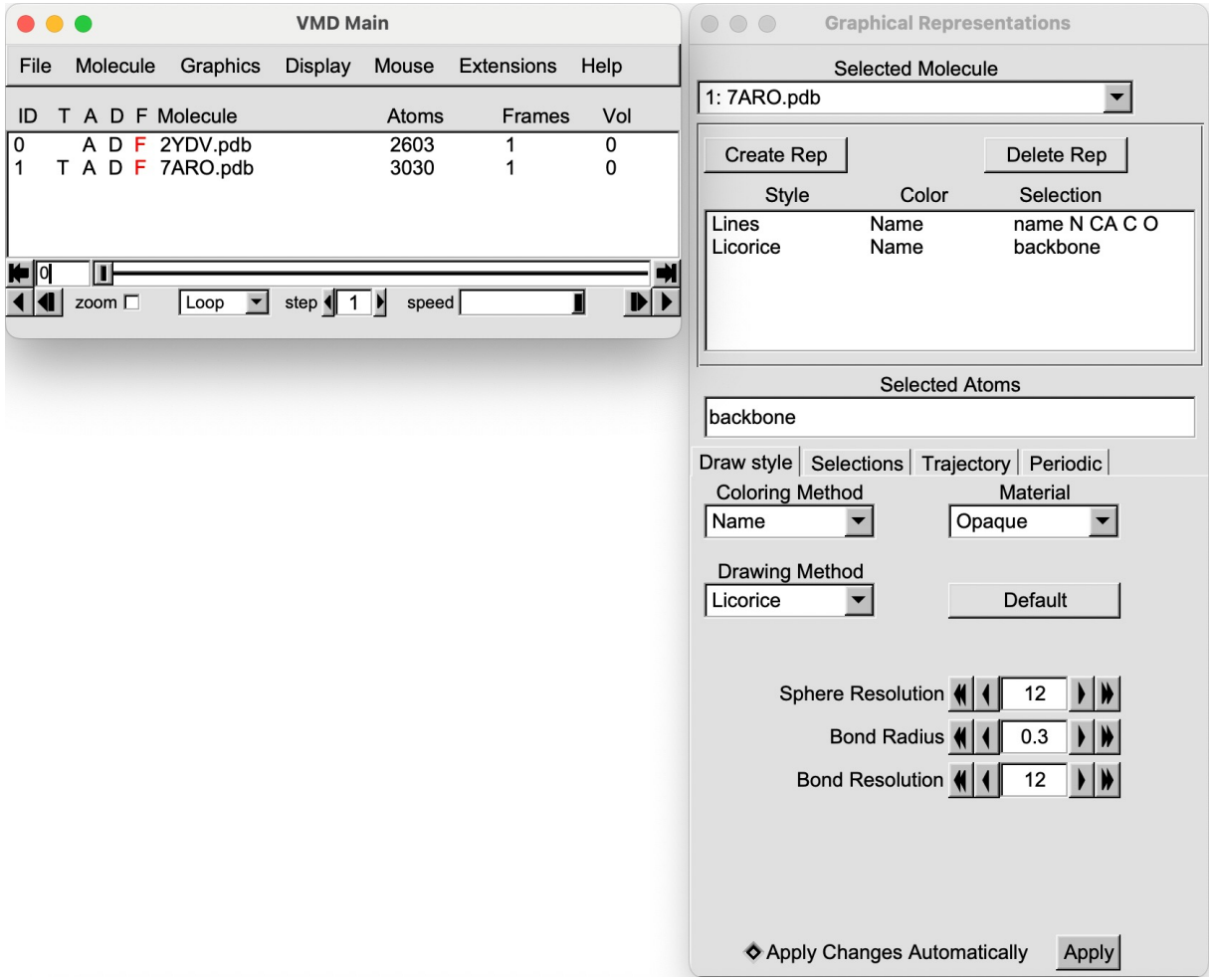
VMD – Interface

- Clone Representations
  - Use the same visualization for all molecules
- Movie Maker
  - Render a trajectory, rotation of a molecule, etc.
- Ruler
  - Display a grid or a ruler



VMD – Representations

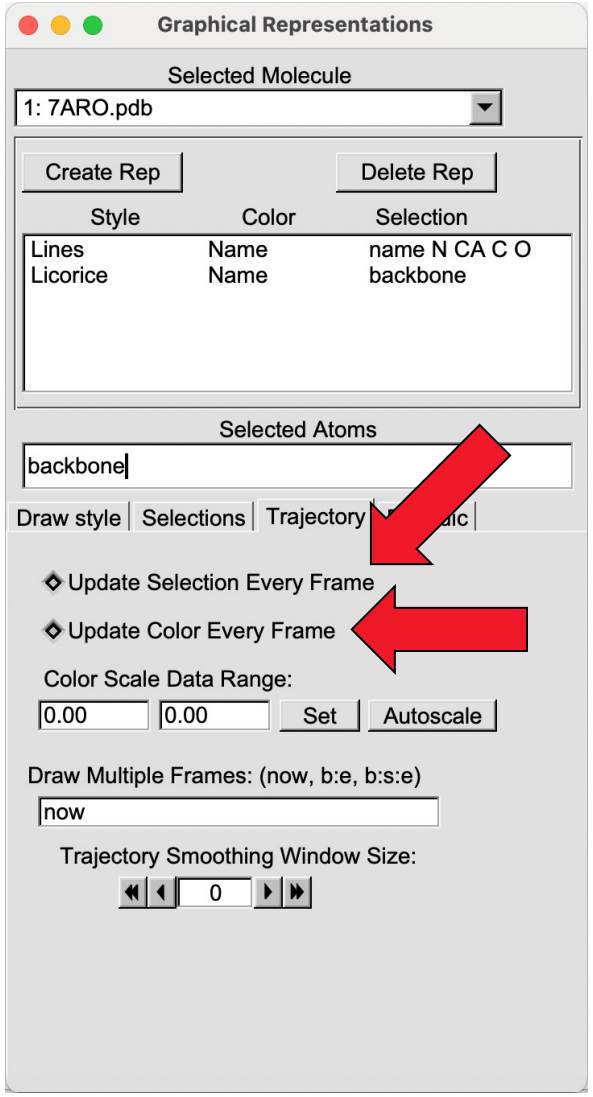
- Representations
  - Multiple representations can be added for each molecule
  - Each representation has a style, coloring method, and is applied on a selection of atoms
  - Each style has one or more parameters that can be adjusted



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VMD – Trajectory

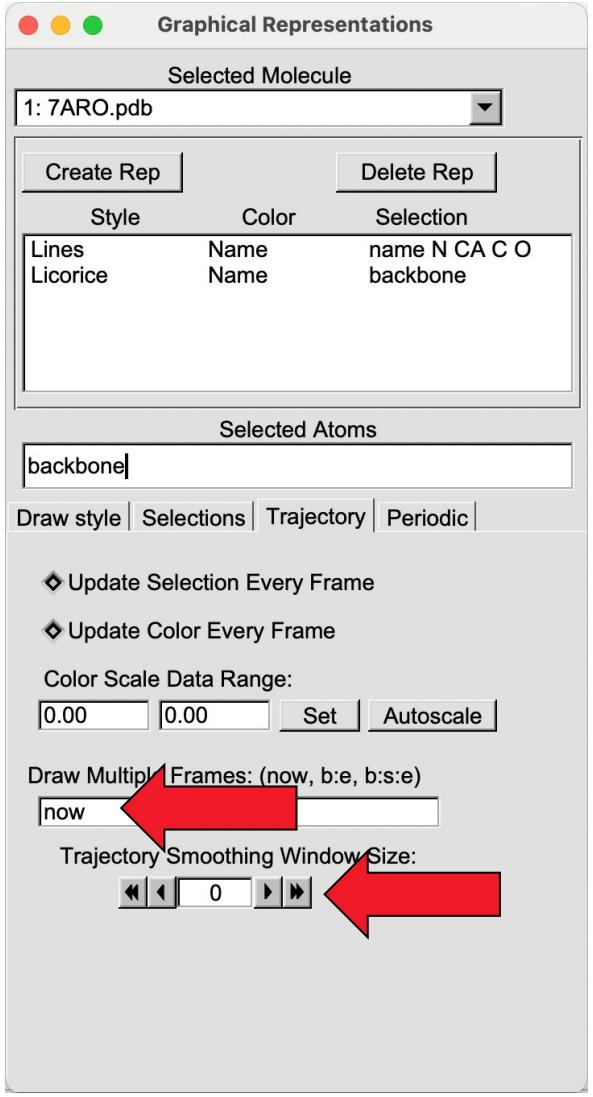
- Update Selection / Color Every Frame
  - Selection of atoms within a certain distance of a reference can change every frame (e.g. name NA CL within 5 of resid 123)
  - Atoms can be colored, e.g., by their coordinates



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VMD – Trajectory

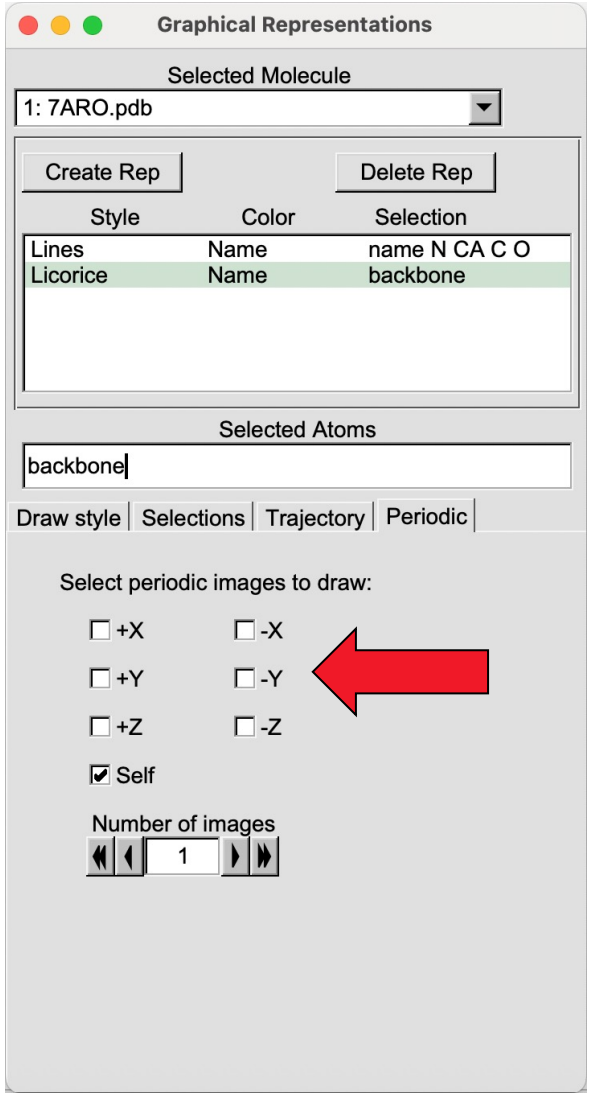
- Draw Multiple Frames
  - Show multiple frames at the same time
  - 0:10:1000 (Every 10th frame in the range of 0 to 1000)
- Trajectory Smoothing
  - Average the atom coordinates over N frames
  - Trajectory becomes less jittery
  - Visualization artifacts with non-physical structures



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VMD – Periodic

- Show multiple periodic images
- Does not work in every “rendermode”
- The unit cell / box dimensions have to be defined
  - CRYST1 record in PDB file
  - Last line in the GRO file

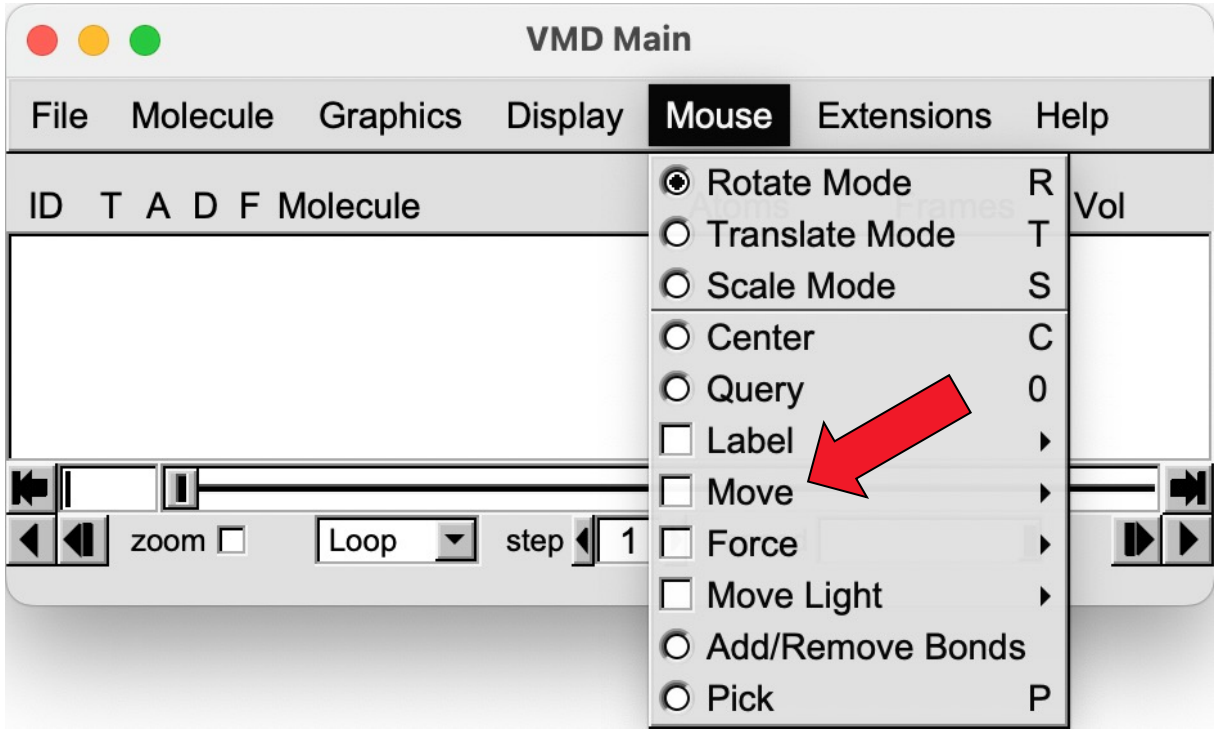


VMD – Tasks

- **Task**
  - Align the trajectory using the CA atoms
  - Use the VolMap Tool with OW atoms as the selection and average over the trajectory
- **Task**
  - Show water molecules within 10 Ångströms of residue 253
  - Pick a water molecule and label the distance from residue 253
  - Plot the distance in time

VMD – Tasks

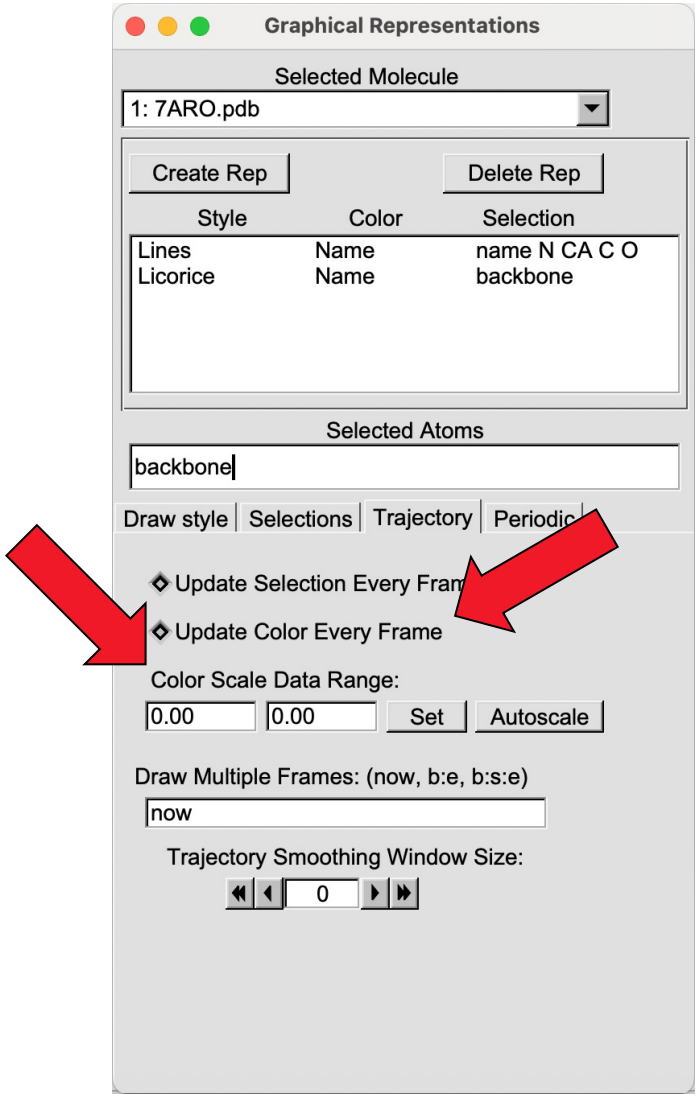
- **Task**
  - Move an atom
  - Move the whole molecule
  - Move a representation



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VMD – Tasks

- **Task**
  - Color atoms by their positions (e.g., radial)
  - Change the color scale range
  - Change the color scale to RGB
  - Try the option to update color every frame



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Useful Commands

- pbc box
- pbc wrap / unwrap
- rotate x by -90
- render ...
- animate goto N
- display resetview
- display projection Orthographic
- display depthcue off
- display ambientocclusion on
- color Display Background white
- axes location Off

Useful Commands

Delete the first representation and add a new one

```
mol delrep 0 top
mol selection {name CA}
mol representation VDW 1.5 25.0
mol color ColorID 3
mol material AOChalky
mol addrep top
```

Loop over trajectory frames

```
set nf [molinfo top get numframes]

for {set i 0} {$i < $nf} {incr i} {
    animate goto $i
    puts ”Frame $i"
}
```

Common Visualization Artifacts

- Caused by automatic bond detection
  - CONNECT records are ignored
- Non-existent bonds
- Missing bonds
- Wrong bond orders
- Periodic boundary conditions – splitting molecules

Common Problems

- PyMOL
  - Reorders atoms when saving a file
- VMD
  - May write invalid files (does not account for the fixed width file format)
  - Takes up too much memory when loading large trajectories